Credit valuation adjustment for derivative contracts

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Agenda

- The project – presentation of key requirements
- Introduction to CVA
- Models for credit exposure
- How to compute the implied probability of default from a CDS spread?
- Q&A

The project – presentation of key requirements

Project scope: Credit valuation adjustment for derivative contracts

Key task: prepare a report describing applied methodology as well as CVA calculation results for different types of derivative instruments

Calculation date: 31.12.2025

The project – presentation of key requirements

Analyzed derivatives:

- Buy EUR sell PLN FX Forward (*maturity: 3 year, notional: 1 000 000 EUR; strike: 4.4439*)
- Receiver (we receive fixed rate) EUR EURIBOR 3M IRS (*maturity: 3 years, notional: 10 000 000 EUR; fixed rate: 2.02% - received annually*)

The project – presentation of key requirements

I. Analysis of derivative instruments in question:

- Mechanics,
- Payoff formula/ payoff profile,
- Valuation formula, can you simplify the formula for the floating leg in the IRS contract?

II. Implied default probability:

- CDS contracts: mechanics,
- CDS contracts: valuation,
- Implying probability of default from a CDS spread: CDS bootstrapping,
- CDS bootstrapping for sample market quotes (**IMPORTANT: algorithm implemented in code form**)

The project – presentation of key requirements

III. Simulation and calculation of expected exposure profiles (EE) for FX Forward:

- Calibration to market data,
- Simulation: 3 years horizon, 1 month time step, minimum number of simulations 10 000,
- Presentation and identification of key features of EE for FX Forward.

IV. Simulation and calculation of expected exposure profiles (EE) for IRS:

- Calibration to market data,
- Simulation: 3 years horizon, 1 month time step, minimum number of simulations 10 000,
- Presentation and identification of key features of EE for IRS,
- <u>Simulation: joint simulation of two processes with correlation assumption included (hint: Cholesky decomposition).</u>

The project – presentation of key requirements

V. CVA calculation for derivatives in question :

- CVA calculation for the analyzed derivatives: analysis both at single instrument as well as at portfolio level (**please present the result at portfolio level in PLN and include the correlation between the FX and IR risk factors**);
- How the CVA at portfolio level is impacted by introduction of the variation margin (VM):
  - VM currency is PLN;
  - VM is exchanged in monthly intervals;
  - VM for time step *T* is calculated using the following algorithm:

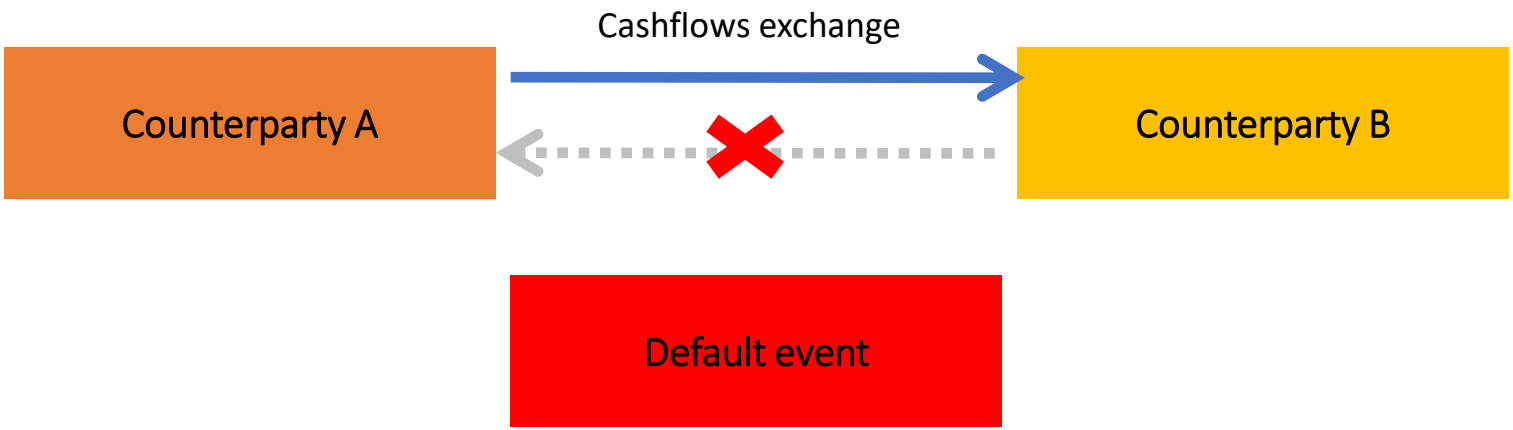
$$VM(T) = MtM_{IRS}(T-1) \cdot FX(T-1) + MtM_{FX\ Forward}(T-1)$$

$$VM(0) = MtM_{IRS}(0) = MtM_{FX\ Forward}(0) = 0$$

Counterparty credit risk

Counterparty risk is the risk to each party of a contract that the counterparty will not live up to its contractual obligations.

Counterparty risk is a risk to both parties and should be considered when evaluating a contract.

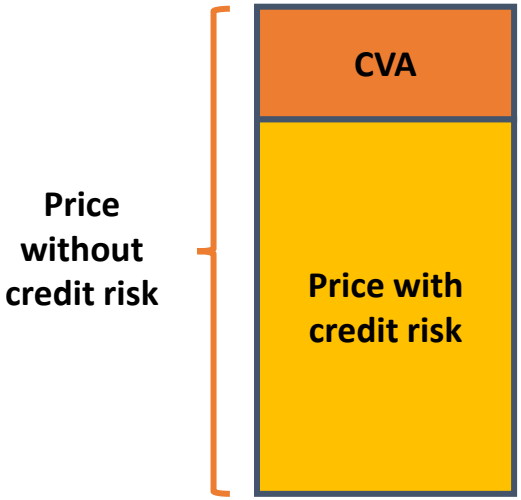


Introduction to CVA

CVA definition

- Counterparty credit risk:
  - The risk that counterparty defaults before the contract matures
  - CVA quantifies counterparty credit risk as a certain add-on to the price of derivative contract

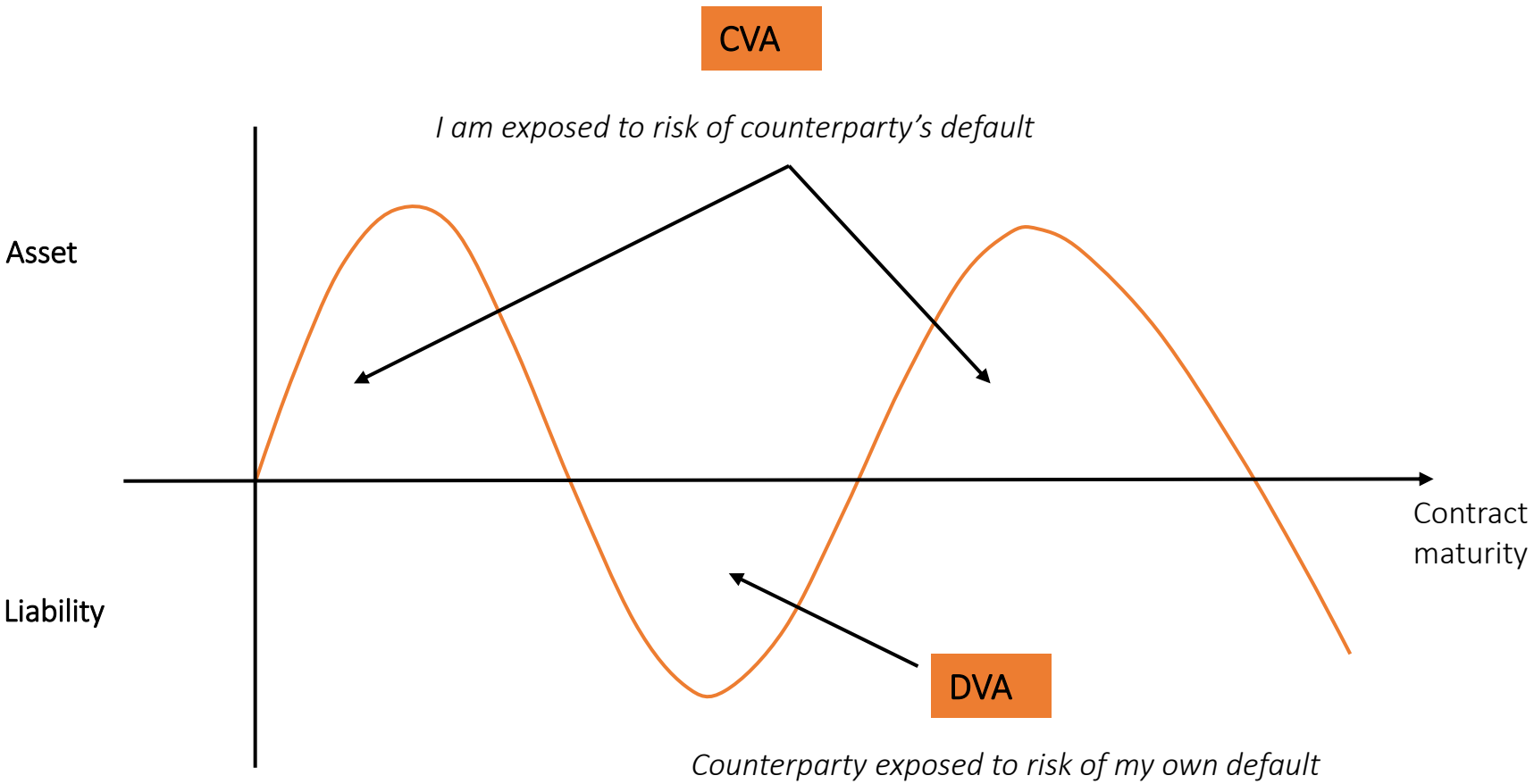
$$\underbrace{\textbf{Price including risk}}_{\text{Credit risk included}} = \underbrace{\textbf{Price}}_{\text{Without credit risk}} - \textbf{CVA}$$



- Two disctinct approaches to CVA:
  - (1) Provision calculated based on historical data
  - (2) Add-on calibrated to current market conditions

Introduction to CVA

The nature of counterparty credit risk exposure



Introduction to CVA

Basic formula for CVA calculation

- Assumptions:
  - No Wrong-Way Risk
  - Correlation between defaults of counterparties omitted

- Unilateral CVA/DVA at a contract level

$$CVA(t) = LGD \int_t^T EE(u) dPD_C(u)$$

where:

- *LGD* – loss given default
- *EE*(*u*) – present value of expected exposure
- $PD_C(u)$ – default probability

Introduction to CVA

CVA calculation in practice

$$CVA \approx (1 - R) \cdot \sum_{i=1}^{N} EE(t_i) \cdot DF(0, t_i) \cdot q(t_{i-1}, t_i),$$

where:

- $EE(t_i)$– expected exposure within t_i,
- $DF(0, t_i)$ – discount rate for the term t_i,
- $q(t_{i-1}, t_i) = PS(t_{i-1}) - PS(t_i)$ – probability of counterparty's default for period from $t_{i-1}$to t_i,
- $PS(t_i)$ – survival probability within t_i,
- R – recovery rate,
- N – Number of analyzed points (grid size).

Models for credit exposure

Geometric Brownian Motion

Stochastic differential equation:

$$dS(t) = S(t) \times \mu(t) \times dt + \sigma \times S(t) \times dW(t),$$

where:

- $S(t)$– the value of underlying at time t,
- $\mu(t)$– the drift,
- σ – the volatility,
- $dW(t)$ - standard Brownian motion.

The GBM process has an exact solution in the time interval $[t_i, t_{i+1}]$:

$$S(t_{i+1}) = S(t_i) \times \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) \times \Delta t + \sigma \times \left(\mathrm{W}(\mathrm{t}_{\mathrm{i}+1}) - \mathrm{W}(\mathrm{t}_{\mathrm{i}})\right)\right].$$

Models for credit exposure

Arithmetic Brownian Motion

Stochastic differential equation:

$$dS(t) = \mu(t) \times dt + \sigma \times dW(t),$$

where:

- $\mu(t)$– the drift,
- σ – the volatility,
- $dW(t)$ - standard Brownian motion.

Models for credit exposure

Vasicek model

Stochastic differential equation:

$$dr(t) = \lambda \times (\mu - r(t)) \times dt + \sigma \times dW(t),$$

where:

- r(t)– the value of short rate at time t,
- μ– the long term mean,
- σ – the volatility,
- $dW(t)$ - standard Brownian motion.

Where to find data for calibration?

FX rate history: EURPLN - Euro / Polish Zloty – Stooq

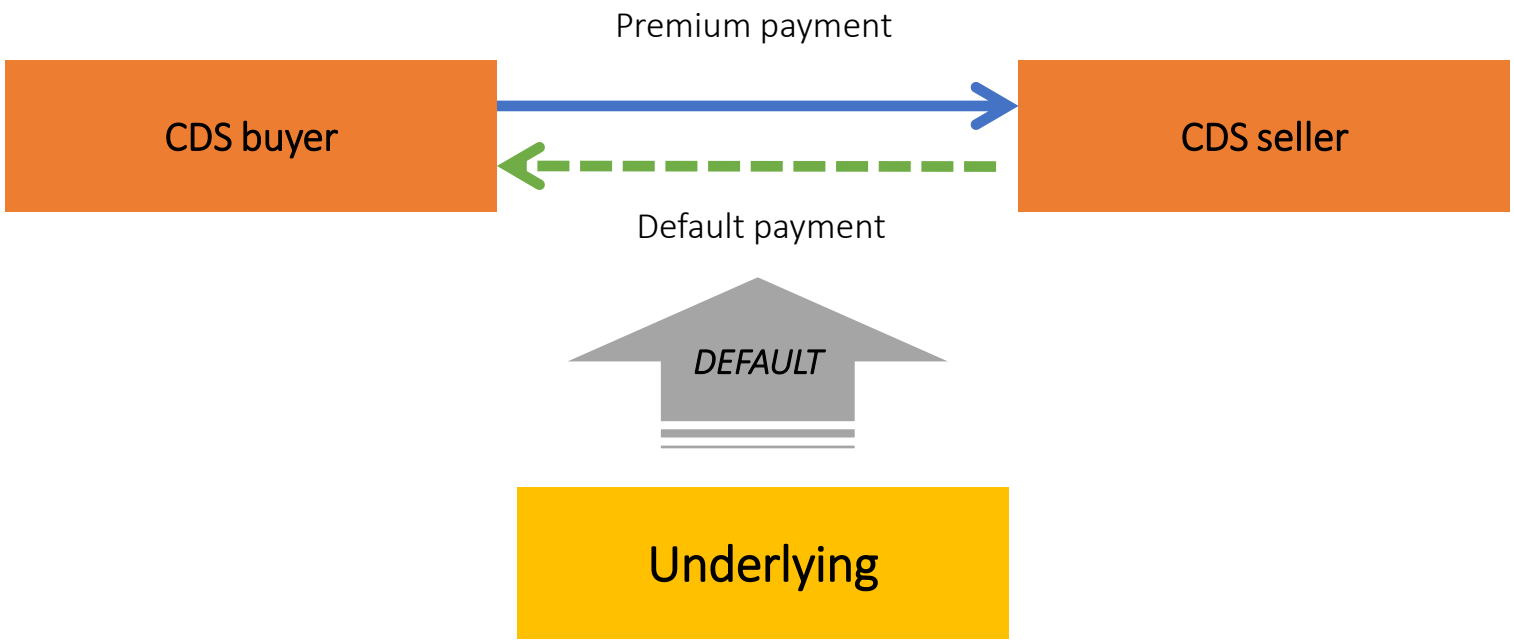
EUR (ESTR) rate history: EST.B.EU000A2X2A25.WT | ECB Data Portal

Sample paper on calibration: Calibration of the Vasicek Model to Historical Data with Python Code

Introduction to CDS

Mechanics

- Cashflows connected with Credit Default Swap transaction



Credit spread

CDS mechanics - Premium Leg

- NPV formula for ***Premium Leg*** (paid by protection buyer) with maturity *N*:

$$PL_N = S_N \cdot \sum_{n=1}^{N} DF(0, T_n) \cdot PS(T_n) \cdot \Delta t_n$$

Parameter	Definition
S_N	CDS spread for contract with maturity *N*
$DF(0,T_n)$	Discount factor (risk free) for period from 0 to T_n
$PS(T_n)$	Survival probability for period from 0 to T_n
Δt_n	Year fraction for period from T_{n-1} to T_n

Credit spread

CDS mechanics - Default Leg

- NPV formula for ***Default Leg*** (paid by protection seller in case of default) with maturity *N*:

$$DL_N = (1 - RR) \cdot \sum_{n=1}^{N} DF(0, T_n) \cdot [PS(T_{n-1}) - PS(T_n)]$$

Parameter	Definition
$DF(0,T_n)$	Discount factor (risk free) for period from 0 to T_n
$PS(T_n)$	Survival probability for period from 0 to T_n
RR	Recovery rate

Credit spread

CDS mechanics – CDS spread

- At the CDS inception NPV of Premium Leg equals NPV of Default Leg:

$$PL_N = DL_N$$

- Based on the above the formula for CDS spread is as follows:

$$S_N = \frac{LGD \cdot \sum_{n=1}^{N} DF(0, T_n)[PS(T_{n-1}) - PS(T_n)\,]}{\sum_{n=1}^{N} DF(0, T_n) PS(T_n) \Delta t_n}$$

Models for credit exposure

CIR model

Stochastic differential equation:

$$d\lambda(t) = \kappa \times (\mu - \lambda(t)) \times dt + \sigma \times \sqrt{\lambda(t)} \times dW(t),$$

where:

- $\lambda(t)$ – the value of hazard rate at time t,
- μ – the long term mean,
- σ – the volatility,
- $dW(t)$ - standard Brownian motion.

Models for credit exposure

CIR model calibration

The relationship between the hazard rate and the survival probability is as follows: $PS(t_i, t_M) = \exp\left(-\int_{t_i}^{t_M} \lambda(s)ds\right)$.

Under the CIR model, at time t_i the survival probability until time t_M can be calculated using the following formula:

$$P(t_i, t_M) = A(t_i, t_M)e^{-B(t_i,t_M)*\lambda(t_i)},$$

where:

$$B(t_i, t_m) = \frac{2(\exp\{(t_m - t_i)h\} - 1)}{2h + (k+h)(\exp\{(t_m - t_i)h\} - 1)},$$

$$A(t_i, t_m) = \left(\frac{2h\exp\left\{\frac{(t_m - t_i)(k+h)}{2}\right\}}{2h + (k+h)(\exp\{(t_m - t_i)h\} - 1)}\right)^{\frac{2\kappa\theta}{\sigma^2}},$$

$$h = \sqrt{\kappa^2 + 2\sigma^2}.$$

Recommended literature

- Jon Gregory, *The xVA Challenge: Counterparty Risk, Funding, Collateral, Capital and Initial Margin 4th Edition*, Wiley Finance, 2020
- https://cvacentral.com/books/
- Cesari, G., Aquilina, J., Charpillon, N., Filipovic, Z., Lee, G., Manda, I., *Modelling, Pricing, and Hedging Counterparty Credit Exposure*, Springer Finance, 2009
- Damiano Brigo, Andrea Pallavicini, *CCPs, Central Clearing, CSA, Credit Collateral and Funding Costs Valuation FAQ: Re-Hypothecation, CVA, Closeout, Netting, WWR, Gap-Risk, Initial and Variation Margins, Multiple Discount Curves, FVA?*, online: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2361697

Thank you